Week 1 Documentation - Data Loading, Cleaning, and Exploration

Project Overview

The objective of this task was to combine multiple product datasets into a single structured dataset and perform initial data preprocessing. The work focused on importing data from several CSV files, exploring the dataset, handling missing values, removing duplicate records, creating useful derived columns, and exporting a clean dataset for future analysis.

Objectives

The main objectives of this task were to:

- Import multiple CSV files into Python.
- Merge all datasets into one DataFrame.
- Explore the dataset to understand its structure.
- Identify and handle missing values.
- Remove duplicate records.
- Perform basic filtering and data selection.
- Convert price values into a numeric format.
- Create additional calculated columns.
- Save the cleaned dataset for further analysis.

Tools and Technologies

Tool	Purpose
Python	Programming language
Pandas	Data manipulation and analysis
OS Module	Working with file paths
Glob	Reading multiple CSV files automatically
Jupyter Notebook	Development environment

Workflow

1. Importing Required Libraries

The notebook begins by importing the required Python libraries:

- Pandas

- OS
- Glob

These libraries are used to read multiple files, manage directories, and perform data analysis efficiently.

2. Loading Multiple CSV Files

Instead of reading each file individually, all CSV files present inside the folder are automatically detected using the glob() function.

Each file is loaded into a temporary DataFrame, after which all DataFrames are combined into one master DataFrame using pd.concat().

This approach is efficient and scalable because new CSV files can be added to the folder without changing the code.

3. Data Exploration

Before cleaning the dataset, an exploratory analysis was performed to understand its contents.

The following operations were carried out:

- Displaying the first five rows
- Displaying the last five rows
- Checking dataset dimensions
- Listing column names
- Viewing data types
- Generating descriptive statistics

This helped verify that the data had been loaded correctly and provided an overview of the dataset.

4. Handling Missing Values

The notebook checks every column for missing values.

Different strategies were used depending on the data type:

Numeric Columns

Missing numerical values were replaced with the column mean to preserve the overall distribution of the data.

Text Columns

Missing text values were replaced with the word **"Unknown"**, ensuring that no null values remained in categorical columns.

5. Data Selection and Filtering

Several filtering operations were performed to better understand the product data.

Examples include:

- Selecting important columns such as title, rating, and final price.
- Displaying products with ratings greater than 4.
- Displaying products offering discounts greater than 50%.
- Identifying products priced above ₹1000.

These operations demonstrate the use of conditional filtering in Pandas.

6. Data Type Conversion

The **Final Price** column originally contained currency symbols and commas, making it unsuitable for numerical calculations.

The cleaning process included:

- Removing the ₹ symbol
- Removing commas
- Converting the values to numeric format

This allowed mathematical operations to be performed on the price column.

7. Duplicate Removal

Duplicate rows were identified using the duplicated() function.

After identifying duplicate records, they were removed using drop_duplicates().

The notebook also verifies that no duplicate records remain after cleaning.

8. Feature Engineering

Two additional columns were created:

Quantity

A default quantity value of **1** was assigned to every product.

Total Amount

A new column named **Total Amount** was calculated using:

Total Amount = Final Price × Quantity

This prepares the dataset for future sales-related analysis.

9. Exporting the Clean Dataset

After completing all preprocessing steps, the cleaned DataFrame was exported as:

cleaned_dataset.csv

A final verification confirmed that the file was successfully saved.

Data Cleaning Techniques Used

- Merging multiple CSV files
- Handling missing values
- Removing duplicate records
- Converting text to numeric values
- Creating calculated columns
- Exporting cleaned data

Outcome

By the end of the notebook:

- All CSV files were successfully merged into one dataset.
- Missing values were handled appropriately.
- Duplicate entries were removed.
- Price values were converted into a usable numeric format.
- Additional calculated fields were created.
- A clean dataset was generated and saved for future analysis and visualization.

Project Summary

This project focused on building a clean and organized dataset from multiple CSV files using Python and Pandas. The workflow involved importing all available files, combining them into a single DataFrame, and performing exploratory data analysis to understand the dataset. Missing values were treated according to their data types, duplicate records were removed, and the **Final Price** column was converted from text to a numeric format for accurate calculations. Basic filtering operations were performed to identify highly rated, heavily discounted, and premium-priced products. Additional fields such as **Quantity** and **Total Amount** were created to support future business analysis. Finally, the processed dataset was exported as a new CSV file, providing a reliable foundation for further analytics, visualization, and machine learning tasks.